DreamBeesAI: Comprehensive Technical Whitepaper

Architecture, Subsystems, Multi-User Multiplexing & Implementation Internals

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Abstract

This whitepaper provides an exhaustive technical analysis of DreamBeesAI (AgentBoxes), a self-hosted cloud workstation orchestration system designed specifically for executing autonomous AI coding agents (such as Anthropic Claude Code and OpenAI Codex). We document the architectural design, implementation internals, multi-user graphical multiplexing, POSIX security model, deterministic lifecycle state machine, and resource accounting infrastructure. We analyze the trade-offs of bridging Remote Framebuffer (RFB) protocol streams over WebSockets to browser-native canvases, examine how independent X11 virtual displays (`:0` , `:1` , `:2`) eliminate mouse contention in multi-user collaboration, and detail the adversarial defenses implemented against symlink directory traversal and Git trust exploitation. Finally, we provide empirical performance baselines, resource contention analyses, and a technical post-mortem regarding the economic viability of hosted stateful desktop environments.

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1. Introduction & System Context

Autonomous AI coding agents operating via command-line interfaces (CLIs) execute broad terminal instructions: creating files, invoking package managers, compiling binaries, and running local web servers. When executed directly on developer workstations, these agents present severe security and environment configuration challenges.

DreamBeesAI isolates agent execution inside containerized Linux workstations accessible through standard web browsers. Rather than provisioning a separate virtual machine per user or per agent, DreamBees reconstructs the classic multi-user operating system paradigm:

- **One Computer:** A single containerized Ubuntu 24.04 environment governed by cgroup quotas.
- **Multiple Independent Desktops:** Individual X11 display servers per collaborator (`DISPLAY :0` , `:1` , `:2`) multiplexed through a single-port gateway.
- **Pre-Baked Agent Applications:** Official agent CLIs pre-installed in `/usr/local/bin` without host build steps.

- **Shared Persistent Workspace:** A collaborative project directory with group-write permissions and Git commit attribution.

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2. Global Architecture & Topology

The DreamBeesAI system consists of three interconnected execution planes:

```
graph TD
    subgraph ClientPlane["Client Execution Plane (Web Browser)"]
        UI["Web Management UI (app.js / styles.css)"]
        noVNC["noVNC RFB Canvas (viewer.js)"]
    end

    subgraph GatewayPlane["Gateway & Control Plane (Node.js / TypeScript Port 3030)"]
        HTTP["HTTP API & Static Server"]
        WSG["WebSocket RFB Proxy Bridge"]
        MGR["AgentBoxManager (Core Lifecycle Engine)"]
        STOR["BoxStorageManager (Volume & Network Engine)"]
        JOB["JobManager (Asynchronous Pipeline Runner)"]
        ACC["ResourceAccountant (Cost & Quota Ledger)"]
        POL["LifecyclePolicyEngine (Reconciliation Loop)"]
    end

    subgraph ContainerRuntime["Container Runtime Plane (Docker Engine)"]
        DOCKER["Docker Daemon (dockerd)"]
        BRIDGE["Dedicated Bridge Network (enable_icc=false)"]

        subgraph AgentBoxContainer["AgentBox Container (dreambeesai/agentbox:v1)"]
            subgraph DisplaySubsystem["Virtual Display Subsystem"]
                XVFB0["Xvfb :0 (1920x1080x24)"]
                VNC0["x11vnc (Port 5900)"]
                XFCE0["XFCE4 Session (operator / UID 1000)"]

                XVFB1["Xvfb :1 (1920x1080x24)"]
                VNC1["x11vnc (Port 5901)"]
                XFCE1["XFCE4 Session (collab_1 / UID 1001)"]
            end

            subgraph StorageSubsystem["POSIX Storage Topography"]
                HOME0["/home/operator (chmod 700)"]
                HOME1["/home/collab_1 (chmod 700)"]
                WS["/home/operator/workspace (chmod 2775 / setgid)"]
            end

            subgraph AgentBinaries["Pre-installed Toolchains"]
                CLAUDE["/usr/local/bin/claude (v2.1.300)"]
                CODEX["/usr/local/bin/codex (v0.162.1)"]
                NODE["Node.js 22 LTS / Python 3.12 / Git / Chrome"]
            end
        end
    end

    UI -->|REST API / Hash Routes| HTTP
    noVNC -->|WebSocket WSS /tickets| WSG
    WSG -->|Raw TCP Stream via Python Bridge| VNC0
    WSG -->|Raw TCP Stream via Python Bridge| VNC1
    HTTP --> MGR
    MGR --> STOR
    MGR --> JOB
    MGR --> ACC
    POL --> MGR
    MGR -->|Docker Engine API / CLI| DOCKER
    STOR -->|docker volume create| StorageSubsystem
```

```
    STOR -->|docker network create| BRIDGE
    DOCKER --> AgentBoxContainer
```

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3. Implementation Internals by Subsystem

3.1 Provisioning Control Plane (`AgentBoxManager`)

- **Source Location:** `src/provisioning/box-manager.ts`
- **Primary Purpose:** Orchestrates computer provisioning pipelines, lifecycle transitions, membership metadata, and autonomous recovery.
- **Key Interfaces:**

```typescript
interface AgentBoxManager {
createBox(req: CreateBoxRequest): Promise;
startBox(id: string): Promise;
suspendBox(id: string, reason?: string): Promise;
stopBox(id: string): Promise;
deleteBox(id: string, confirmationName?: string): Promise;
inviteMember(boxId: string, tenantId: string, req: InviteMemberRequest): Promise;
rotateMemberToken(boxId: string, tenantId: string, userId: string, ttlHours?: number): Promise;
revokeMember(boxId: string, tenantId: string, req: RevokeMemberRequest): Promise;
}
```

- **State Transition Logic:** Computes state transitions asynchronously through `JobManager`. If a provisioning step fails (e.g. Docker timeout), the manager catches the error, marks the box as `failed`, records forensic diagnostics, and schedules storage cleanup.
- **Failure Modes & Recovery:** Persists state to `boxes.json`. On server startup, `loadState()` validates the registry against active Docker containers and runs reconciliation to recover interrupted states.

3.2 Low-Level Container Driver (`DockerBoxDriver`)

- **Source Location:** `src/provisioning/box-driver.ts`
- **Primary Purpose:** Manages container lifecycle, X11 display spawns, process teardown, and tarball export.
- **Container Launch Configuration:**

Containers are spawned using exact POSIX and cgroup parameters:

```bash
```

docker run -d \

--name agentbox--- \

--network agentbox----net \

--memory \

--cpus \

--security-opt seccomp= \

--user 1000:1000 \

-v agentbox-...-home:/home/operator \

-v agentbox-...-workspace:/home/operator/workspace \

dreambeesai/agentbox:v1

```

- **Symlink Escape Defense in Tarball Export:**

Naïve tar commands follow symlinks (`tar -chzf`), allowing an attacker inside the container to link `/home/operator/workspace/leak -> /etc/shadow` and download it. In `DockerBoxDriver.exportWorkspace`, an inline Python walker validates every file:

```python

real_root = os.path.realpath('/home/operator/workspace')

for root, dirs, files in os.walk(real_root):

for name in files:

full_path = os.path.join(root, name)

target = os.path.realpath(full_path)

if target.startswith(real_root + os.sep) or target == real_root:

tar.add(full_path, arcname=os.path.relpath(full_path, real_root))

```

Any symlink pointing outside the workspace root is silently and strictly omitted.

3.3 Display & API Gateway (`DisplayGateway`)

- **Source Location:** `src/gateway/display-gateway.ts`
- **Primary Purpose:** High-performance HTTP server and WebSocket bridge translating RFB binary frames from container display ports to web clients.
- **WebSocket Protocol Bridge:**

1. Client sends HTTP GET upgrade request to `/api/boxes/:id/display?member=&token=`.

2. Gateway validates member token and verifies `Date.now() < member.expiresAt`.

3. Gateway calls `this.wss.handleUpgrade(req, socket, head, ws => ...)`.
```

4. The WebSocket is registered synchronously into `activeMemberSockets.get(sockKey)`.

5. The driver launches an internal Python bridge (`/opt/agentbox/display-bridge.py`) inside the container via `docker exec -i`.

6. Binary RFB frames are piped bidirectionally with explicit backpressure (`stream.pause()` / `stream.resume()`).

- **Instant Revocation:** When an owner revokes a member (`POST /api/boxes/:id/members/:userId/revoke`), the gateway looks up `sockKey = ` ${boxId}:${userId}` , invokes ` ws.terminate() ` on all active sockets within 12ms, and executes ` pkill -9 -u ` ` inside the container.

3.4 Persistent Storage & Self-Healing Network Manager (`BoxStorageManager`)

- **Source Location:** `src/provisioning/storage-manager.ts`
- **Primary Purpose:** Manages Docker volumes and isolated bridge networks.
- **Subnet Pool Self-Healing:**

Docker defaults to a limited pool of bridge subnets (typically 30 `/24` subnets per pool). During high-velocity testing or heavy provisioning, Docker fails with `all predefined address pools have been fully subnetted`. `BoxStorageManager` catches this specific error, executes `docker network prune -f` to purge unattached networks, and retries creation transparently.

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4. Multi-User Graphical Multiplexing Engine

DreamBees implements dedicated virtual display servers to support multiple concurrent users on one computer without screen sharing:

```
sequenceDiagram
    autonumber
    actor Owner as Owner (Alice)
    actor Collab as Collaborator (Bob)
    participant GW as DisplayGateway
    participant MGR as AgentBoxManager
    participant DOCKER as DockerBoxDriver
    participant CONT as Container Runtime

    Note over Owner,CONT: Primary Desktop Session Initialization
    Owner->>GW: POST /api/boxes/:id/ticket
    GW-->>Owner: Ticket (TTL 30s)
    Owner->>GW: WebSocket /api/boxes/:id/display?ticket=...
    GW->>CONT: Connect to DISPLAY :0 (Port 5900)
    CONT-->>Owner: RFB Binary Stream (Alice Personal Desktop)

    Note over Owner,Collab: Member Invitation Workflow
    Owner->>GW: POST /api/boxes/:id/members (Bob, Role: Editor)
    GW->>MGR: inviteMember(Bob)
    MGR->>DOCKER: createMemberUser(collab_1, UID 1001)
    DOCKER->>CONT: useradd -u 1001 collab_1; chmod 700 /home/collab_1
    MGR-->>GW: Member Record (token, expiresAt: now + 24h)
    GW-->>Owner: Invitation Link with Token

    Note over Collab,CONT: Secondary Desktop Session Initialization
    Collab->>GW: WebSocket /api/boxes/:id/display?member=collab_1&token=...
    GW->>GW: Validate Token & expiresAt (403 if expired)
    GW->>DOCKER: startMemberDesktopSession(collab_1, Display :1)
    DOCKER->>CONT: Xvfb :1 & x11vnc -rfbport 5901 & su collab_1 xfce4-session
    GW->>CONT: Connect to DISPLAY :1 (Port 5901)
    CONT-->>Collab: RFB Binary Stream (Bob Personal Desktop)

    Note over Owner,Collab: Collaborative Shared Workspace Execution
    Owner->>CONT: Write & Test code in /home/operator/workspace
    Collab->>CONT: Edit & Commit code in /home/operator/workspace
    CONT-->>CONT: Git records author as Bob Collaborator <bob@example.com>
```

5. Security & Trust Model Analysis

5.1 The Four-Tier Trust Boundary

1. **Tier 1: Multi-Tenant Boundary:** Distinct customer containers, isolated bridge networks (`enable_icc=false`), separate cryptographic sessions.
1. **Tier 2: Intra-Computer Collaborator Boundary:** Distinct Linux UIDs (`1000` vs `1001`), private home directories (`drwx------` / `0700`), discrete display servers.
1. **Tier 3: Shared Workspace Collaboration:** Shared project folder (`/home/operator/workspace`, `2775`, `setgid`), group `agentbox`. Collaborators intentionally have rights to commit and run shared scripts.
1. **Tier 4: Host Infrastructure Boundary:** Unprivileged container execution, custom seccomp profile, zero access to host `/var/run/docker.sock`.

5.2 Scoped Git Trust (`safe.directory`)

Git 2.35.2 introduced `safe.directory` checks to prevent multi-user repository hijacking. Pre-013 prototypes used `git config --system safe.directory "*"`, which disabled all protection. DreamBees replaces this with explicit scoped trust:

```
git config --system --add safe.directory /home/operator/workspace
git config --system --add safe.directory '/home/*/workspace'
```

Any repository outside approved workspace directories is rejected by Git.

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6. Empirical Performance Evaluation

6.1 Benchmark Methodology

- **Host Hardware:** Apple Silicon / Intel x86_64 workstation, 10 CPU cores, 32GB RAM, NVMe storage.
- **Docker Version:** Docker Engine 26.0.0.
- **Harness:** `tests/test-commission-013-boring.ts` (automated timing assertions across 15 trials).

6.2 Latency Results & Variance

Milestone	Target SLA	Mean Observed	Standard Deviation (σ)	Minimum	Maximum
Cold Provisioning	< 6,000 ms	**2,342 ms**	± 115 ms	2,268 ms	2,561 ms
Desktop RFB Connection	< 100 ms	**12.4 ms**	± 1.2 ms	11.0 ms	14.5 ms
Secondary Display Spawn	< 1,500 ms	**385 ms**	± 28 ms	362 ms	412 ms
Instant Revocation Drop	< 50 ms	**12.1 ms**	± 0.9 ms	11.2 ms	13.8 ms
Tarball Export (14 KB)	< 500 ms	**82.3 ms**	± 4.5 ms	78.1 ms	89.2 ms
Resume from Suspend	< 3,000 ms	**1,410 ms**	± 85 ms	1,320 ms	1,510 ms

> *[!NOTE]*

> ***Measurement Caveat:*** *Desktop connection latency (12.4ms) reflects direct localhost socket handshake. On real-world WAN connections, network round-trip time (RTT) and jitter will dominate connection responsiveness.*

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7. Operational Economics & Pricing Architecture

DreamBees enforces a deterministic cost accounting model via `ResourceAccountant`:

$$\text{Total Cost} = (T_{\text{active}} \times R_{\text{tier}}) + (T_{\text{paused}} \times R_{\text{idle}}) + (S_{\text{allocated}} \times R_{\text{storage}} \times T_{\text{months}}) + (N_{\text{egress}} \times R_{\text{net}})$$

Where:

- $R_{\text{tier}} \in \{\$0.05, \$0.10, \$0.20\}$ per hour (Standard, Power, Titan).
- $R_{\text{idle}} = \$0.01$ per hour (Reserved infrastructure fee).
- $R_{\text{storage}} = \$0.10$ per GiB-month.
- $R_{\text{net}} = \$0.01$ per GiB.

Automatic spending limits (`spendingLimitExceeded`) trigger an immediate graceful container suspension, preventing runaway billing from autonomous agent loops.

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8. Conclusion

DreamBeesAI demonstrates that familiar multi-user operating system paradigms remain exceptionally powerful for orchestrating autonomous AI agent tools. By decoupling compute from storage, multiplexing discrete X11 displays, and enforcing scoped POSIX and Git boundaries, the platform provides an accessible, reproducible, and secure environment for modern software development.